Software Requirements Specification for Digital Document Verification System

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Revision History

Name	Date	Reason For Changes	Version
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Digital Document Verification System (DDVS), Version 1.0. The purpose of this document is to provide a clear, complete, and unambiguous description of the system's intended behavior so that it can serve as a common reference for the development team, project guide/evaluator, and any other stakeholders involved in the design, implementation, and testing of the system.

This document covers the complete set of requirements for the initial release of the system and is intended to be used as the primary basis for system design, implementation, and acceptance testing.

1.2 Document Conventions

This document follows the conventions listed below:

- Functional requirements are identified using the prefix FR- followed by a two-digit sequence number (e.g., FR-01).
- Non-functional requirements are identified using the prefix NFR- followed by a two-digit sequence number (e.g., NFR-01).
- The keyword "shall" indicates a mandatory requirement. The keyword "should" indicates a desirable but non-mandatory characteristic.
- Section headings and numbering follow the IEEE 830-based SRS template provided for this project.
- Terms defined in Appendix A (Glossary) are used consistently throughout this document.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audience:

- Project team members (developers), who will use this document as the basis for system design and implementation.
- Project guide / evaluator, who will use this document to assess the scope and completeness of the project.
- Testers, who will use the functional and non-functional requirements to design test cases.
- Future maintainers of the system, who may refer to this document to understand original design intent.

Readers who wish to understand the overall purpose and scope of the system should begin with Section 1 and Section 2. Readers primarily interested in specific system behavior should refer to Section 3 (External Interface Requirements) and Section 4 (System Features). Readers interested in quality attributes and constraints should refer to Section 5.

1.4 Product Scope

Digital documents such as certificates, mark sheets, offer letters, and identity proofs are increasingly shared and stored electronically. Because digital files can be copied and modified with common software tools, a recipient of such a document (for example, an employer, an educational institution, or a government office) often has no simple way to confirm whether the document is genuine or has been altered after it was originally issued.

The Digital Document Verification System (DDVS) is a web-based application that allows authorized organizations to register/issue digital documents against unique verification records, and allows any verifier to check the authenticity of a document using a verification ID, a QR code, or by uploading the document itself. The system compares the submitted document or ID against stored records and returns a verification result indicating whether the document is Verified, Invalid, Not Found, or Tampered.

The key objectives of the system are to:

- Provide a simple and reliable mechanism for verifying the authenticity of digital documents.
- Allow authorized organizations to register documents and generate verification records.
- Reduce the effort required by verifiers to manually confirm document authenticity.
- Maintain a record of verification attempts for audit purposes.

The system is designed as an academic, college-level Software Engineering project. It does not use blockchain, artificial intelligence, optical character recognition, or biometric technologies; document integrity is verified using standard cryptographic hashing and record comparison against a relational database.

1.5 References

The following references were used or are relevant to the preparation of this document. Items marked (to verify) should be checked by the author for exact publication details before final submission.

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications, IEEE Computer Society. (to verify edition/year against institution library).
- Sommerville, I., Software Engineering, Pearson Education (to verify edition).
- OWASP Foundation, OWASP Top Ten Web Application Security Risks, available at owasp.org (to verify current version at time of submission).
- RFC 2818, HTTP Over TLS, Internet Engineering Task Force (to verify).
- National Institute of Standards and Technology (NIST), Secure Hash Standard (SHS), FIPS PUB 180-4 (to verify).

Additional references specific to the technology stack chosen for implementation (e.g., framework or database documentation) should be added here once the implementation platform is finalized.

2. Overall Description

2.1 Product Perspective

The Digital Document Verification System is a new, self-contained web-based application. It is not a modification of an existing product and does not replace any legacy system currently in use by the institution or organization for which it is being developed.

The system follows a standard three-tier web architecture consisting of a client-side user interface (accessed through a web browser), a server-side application that implements business logic and processes requests, and a backend relational database that stores user accounts, document records, and verification logs. The system communicates with clients over HTTPS and, where applicable, exposes a REST-based API layer that the front-end consumes.

The system is intended to operate as a standalone application. It does not depend on any external verification authority, government database, or third-party document registry; all documents recognized as "verified" are those explicitly registered within the system by an authorized issuing organization.

2.2 Product Functions

At a high level, the system provides the following major functions:

- Registration and authentication of users (verifiers, issuing organizations, and administrators).
- Registration/issuance of digital documents by authorized organizations, including generation of a unique verification ID and an integrity hash.
- Verification of a document's authenticity using a verification ID, QR code, or an uploaded copy of the document.
- Detection of tampering by comparing the integrity hash of a submitted document with the stored hash of the registered document.
- Maintenance of a verification history log, recording each verification attempt.
- Administrative management of users, registered documents, and system activity.

A summarized function-to-user-class mapping is provided in the table below.

Function	General User / Verifier	Issuing Organization	Administrator
Register and log in	Yes	Yes	Yes (pre-provisioned)
Register/issue a document	No	Yes	No
Verify a document	Yes	Yes	Yes
View own verification history	Yes	Yes	Yes
View system-wide verification activity	No	No	Yes
Manage users	No	No	Yes

Function	General User / Verifier	Issuing Organization	Administrator
Disable/remove document records	No	Limited (own records)	Yes

2.3 User Classes and Characteristics

The system supports the following three user classes.

General User / Verifier

This class includes any individual or organization that wants to check whether a document is genuine (for example, an employer verifying a candidate's certificate). Verifiers are expected to have basic computer literacy and familiarity with using a web browser. No specialized technical knowledge is required. A verifier may optionally register an account to view their own past verification history, or may use the verification feature without registering, depending on the access policy configured for the deployment.

Authorized Issuing Organization

This class includes institutions or organizations (for example, an educational institution issuing mark sheets, or a company issuing offer letters) that are authorized to register documents in the system. Users in this class are expected to have moderate computer literacy. Their accounts are created either through a registration process that is subsequently approved by an administrator, or are directly provisioned by an administrator, depending on the institutional policy adopted.

System Administrator

This class includes personnel responsible for managing the overall system: approving or managing issuing-organization accounts, managing general user accounts, monitoring verification activity, and maintaining the integrity of the document registry. Administrators are expected to have a good understanding of the system's functionality and are the only users granted access to system-wide management functions.

2.4 Operating Environment

The system is a web-based application and is expected to operate in the following environment:

- Client side: Any modern desktop or mobile web browser (e.g., a recent version of Chrome, Firefox, Edge, or Safari) with JavaScript enabled and an active internet connection.
- Server side: A web/application server capable of hosting the server-side application logic (specific technology to be finalized during design, e.g., a Node.js, Java, or Python-based backend).
- Database: A relational database management system (e.g., MySQL or PostgreSQL) used to store user accounts, document records, and verification logs.
- The system does not require any client-side installation; all functionality is accessed through the browser.

2.5 Design and Implementation Constraints

The following constraints apply to the design and implementation of the system:

- The system must be accessible through standard web browsers without requiring browser plug-ins.
- All communication between the client and the server shall use HTTPS to protect data in transit.
- The system shall use a relational database for structured storage of user, document, and verification-log data.
- Document integrity checking shall be performed using a standard cryptographic hash function (e.g., SHA-256) rather than any proprietary or unverifiable method.
- The project is developed as a college-level Software Engineering project; the implementation is expected to run on limited hosting infrastructure (e.g., a single application server and a single database instance) rather than a distributed, high-availability production deployment.
- The system shall not depend on blockchain, artificial intelligence, optical character recognition, or biometric identification technologies.

2.6 User Documentation

The following user-facing documentation is expected to accompany the delivered system:

- An in-application help/FAQ section covering registration, document verification, and document submission procedures.
- A short user guide (delivered as a separate document or as an in-app onboarding sequence) describing how to register an account, verify a document, and interpret verification results.
- An administrator guide describing account approval, user management, and document-record management procedures.

Documentation will be made available electronically (e.g., as a PDF or a set of in-app help pages) rather than in printed form.

2.7 Assumptions and Dependencies

The requirements described in this document rely on the following assumptions:

- Users have access to a device with a modern web browser and a working internet connection.
- Issuing organizations are responsible for the accuracy of the document metadata they submit at the time of registration; the system verifies consistency with the registered record but cannot independently confirm the real-world accuracy of the original document content.
- Administrator accounts are provisioned through a secure, out-of-band process (e.g., manual account creation) prior to system go-live and are not created through public self-registration.
- The system assumes that a document is genuine at the time of registration; the system's role is limited to detecting subsequent modification, not verifying the original authenticity of content before registration.
- Hosting infrastructure (server and database) is assumed to be provisioned and maintained outside the scope of this SRS.

3. External Interface Requirements

3.1 User Interfaces

The system provides a web-based graphical user interface consisting of the following primary screens:

- Login and Registration pages, allowing users to create an account or sign in.
- User Dashboard, providing a verifier or issuing-organization user with access to relevant functions (document verification, document registration for issuers, and personal verification history).
- Document Registration interface, used by authorized issuing organizations to submit document details and register a document.
- Document Verification interface, used to submit a verification ID, scan/enter a QR code value, or upload a document for verification.
- Verification Result page, displaying the outcome of a verification request (Verified, Invalid, Not Found, or Tampered) along with relevant document metadata when applicable.
- Admin Dashboard, providing system administrators with access to user management, document-record management, and verification-activity views.

General usability requirements: the interface shall use consistent navigation and terminology across all pages, shall clearly indicate mandatory input fields, shall display clear error messages for invalid input, and shall be usable on both desktop and common mobile screen sizes.

3.2 Hardware Interfaces

The system does not require any specialized or dedicated hardware. It is accessed using standard end-user hardware, specifically:

- A desktop, laptop, or mobile device capable of running a modern web browser.
- A standard input device (keyboard, mouse, or touchscreen) for entering verification IDs and uploading documents.
- Optionally, a device camera (on desktop or mobile) may be used by the browser to scan a printed QR code, where this feature is supported by the browser and device.

No specialized scanning hardware, biometric hardware, or other dedicated equipment is required.

3.3 Software Interfaces

Software	Purpose / Interface Description
Web Browser (client)	Renders the user interface and communicates with the server over HTTPS using standard HTTP requests.
Application Server	Hosts the server-side business logic, including authentication, document registration, and verification processing.
Relational Database (e.g., MySQL/PostgreSQL)	Stores user accounts, document records, integrity hashes, and verification history logs. Accessed by the application server through standard database drivers/queries.

Software	Purpose / Interface Description
Operating System (server)	Provides the runtime environment for the application server and database (e.g., a Linux-based server OS).

3.4 Communications Interfaces

The system uses the following communication interfaces:

- HTTPS (HTTP over TLS) shall be used for all communication between the client browser and the application server, to protect login credentials and document data in transit.
- Where the front-end and back-end are implemented as separate components, communication between them shall occur through a REST-based API using JSON as the data exchange format.
- The application server communicates with the database using the standard connection protocol/driver of the chosen database management system, over a network connection restricted to trusted hosts.
- No email, SMS, or third-party messaging integration is required for the core scope of this system; if implemented, email notifications (e.g., for account approval) would use standard SMTP and are noted as an optional enhancement rather than a core requirement.

4. System Features

This section describes the major features of the Digital Document Verification System. For each feature, a description and priority, typical stimulus/response sequences, and detailed functional requirements are provided.

4.1 User Authentication and Account Management

4.1.1 Description and Priority

This feature allows individuals and organizations to create accounts, log in, and access functionality appropriate to their assigned role (General User/Verifier, Issuing Organization, or Administrator). It also covers password handling and session management. Priority: High — this feature is a prerequisite for all role-restricted functionality in the system.

4.1.2 Stimulus/Response Sequences

- User submits the registration form with name, email, and password → System validates the input, checks for a duplicate email, creates the account, and displays a confirmation message.
- User submits login credentials → System validates the credentials against stored records and either grants access with a role-appropriate dashboard or displays an error message.
- Issuing-organization user submits a registration request → System stores the request in a pending state → Administrator reviews and approves or rejects the request → System notifies the outcome on the requester's next login.
- User selects "Log out" → System terminates the active session and redirects to the login page.

4.1.3 Functional Requirements

FR-01: The system shall allow a new general user to register by providing a name, a valid email address, and a password.

FR-02: The system shall reject registration attempts that use an email address already associated with an existing account and shall display an appropriate error message.

FR-03: The system shall require passwords to meet a minimum complexity policy (minimum 8 characters, including at least one letter and one number) at the time of registration or password change.

FR-04: The system shall store passwords only in a securely hashed form and shall never store or display passwords in plain text.

FR-05: The system shall allow a registered user to log in using a valid email address and password.

FR-06: The system shall display a generic error message on failed login attempts without indicating whether the email address or the password was incorrect.

FR-07: The system shall allow an organization to submit a request for an Issuing Organization account, which shall remain in a "Pending" state until reviewed by an Administrator.

FR-08: The system shall restrict access to document-registration functionality to users with an approved Issuing Organization role.

FR-09: The system shall restrict access to administrative functionality to users with an Administrator role.

FR-10: The system shall allow a logged-in user to log out, terminating the active session.

FR-11: The system shall automatically terminate an inactive session after a configurable period of inactivity (default: 30 minutes).

FR-12: The system shall allow a logged-in user to change their password after re-entering their current password.

4.2 Document Registration / Issuance

4.2.1 Description and Priority

This feature allows an authorized Issuing Organization to register a digital document within the system. Registration stores the document's metadata and a cryptographic integrity hash of the document file, and generates a unique verification ID (and, optionally, an associated QR code) that can later be used to verify the document. Priority: High — without this feature, no documents exist in the system for verification.

4.2.2 Stimulus/Response Sequences

- Issuing-organization user selects "Register New Document" and enters document metadata (e.g., document type, recipient name, issue date) and uploads the document file → System validates the input, computes an integrity hash of the file, generates a unique verification ID, and stores the record.
- System generates a unique verification ID → System optionally generates a corresponding QR code encoding the verification ID → System displays the verification ID and QR code to the issuing-organization user for distribution with the document.
- Issuing-organization user views their list of previously registered documents → System displays a paginated list of the organization's document records.

4.2.3 Functional Requirements

FR-13: The system shall allow an authenticated Issuing Organization user to register a new document by submitting required metadata (at minimum: document type, recipient name, and issue date) and the document file.

FR-14: The system shall reject document-registration submissions with missing mandatory metadata fields and shall indicate which fields are missing.

FR-15: The system shall compute a cryptographic hash (e.g., SHA-256) of each uploaded document file at the time of registration and store the hash together with the document record.

FR-16: The system shall generate a unique verification ID for each registered document, and this ID shall not be reused for any other document record.

FR-17: The system shall be able to generate a QR code that encodes the verification ID for a registered document.

FR-18: The system shall associate each registered document record with the Issuing Organization account that submitted it.

FR-19: The system shall allow an Issuing Organization user to view a list of documents they have registered.

FR-20: The system shall restrict a document-registration request to file types explicitly supported by the system (e.g., PDF) and shall reject unsupported file types with an appropriate error message.

FR-21: The system shall limit the maximum file size accepted for document upload to a configurable limit (default: 10 MB) and shall reject files exceeding this limit.

4.3 Document Verification

4.3.1 Description and Priority

This feature allows any user (verifier) to check the authenticity of a document by entering its verification ID, scanning/entering a QR code value, or uploading a copy of the document. The system looks up the corresponding record and returns a verification result. Priority: High — this is the core, user-facing purpose of the system.

4.3.2 Stimulus/Response Sequences

- User enters a verification ID → System validates the input format, searches the database for a matching record, and displays the verification result.
- User scans or enters a QR code value → System decodes the value to a verification ID, searches for a matching record, and displays the verification result.
- User uploads a document file for verification → System computes the hash of the uploaded file, compares it against stored hashes of registered documents, and displays whether a matching, unmatched, or tampered record was found.
- System does not find any record matching the submitted ID → System displays a "Not Found" result.
- System finds a matching record but the uploaded file's hash does not match the stored hash → System displays a "Tampered" result.

4.3.3 Functional Requirements

FR-22: The system shall allow a user to submit a verification request using a verification ID entered manually.

FR-23: The system shall allow a user to submit a verification request by scanning or entering a QR code that encodes a verification ID.

FR-24: The system shall allow a user to submit a verification request by uploading a copy of the document to be checked.

FR-25: The system shall validate the format of a manually entered verification ID before performing a database lookup and shall display an error message for malformed input.

FR-26: When a verification ID does not correspond to any registered document, the system shall display a result of "Not Found" and shall not display any document metadata.

FR-27: When a verification ID corresponds to a registered document and, where a file was submitted, the computed hash matches the stored hash, the system shall display a result of "Verified" along with the relevant non-sensitive document metadata.

FR-28: When a verification ID corresponds to a registered document but the computed hash of an uploaded file does not match the stored hash, the system shall display a result of "Tampered".

FR-29: When a document record has been marked as invalid or revoked by an Issuing Organization or Administrator, the system shall display a result of "Invalid" instead of "Verified", even if the verification ID and hash otherwise match.

FR-30: The system shall complete and display the result of a verification request without requiring the user to be logged in, unless the deployment configuration restricts verification to authenticated users.

FR-31: The system shall not display a "Verified" result for any document ID or file that has not been explicitly registered in the system.

4.4 Document Integrity / Tampering Detection

4.4.1 Description and Priority

This feature underlies the verification process by determining whether a submitted document's content matches the content that was originally registered. It relies on comparison of cryptographic hash values rather than manual visual inspection. Priority: High — this feature is what distinguishes genuine verification from a simple ID lookup.

4.4.2 Stimulus/Response Sequences

- User uploads a document as part of a verification request → System computes the hash of the uploaded file using the same algorithm used at registration time → System retrieves the stored hash for the corresponding record (if any) → System compares the two hash values → System reports whether the document is unmodified (matching hash) or modified (non-matching hash).
- System detects a mismatch between computed and stored hash → System flags the record as a tampering detection event and includes it in the verification history log.

4.4.3 Functional Requirements

FR-32: The system shall compute the integrity hash of an uploaded document using the same hash algorithm that was used when the document was originally registered.

FR-33: The system shall compare the computed hash of a submitted document against the stored hash of the corresponding registered record when a document file is provided as part of a verification request.

FR-34: The system shall flag a verification attempt as "Tampered" whenever the computed hash does not match the stored hash for an otherwise matching verification ID.

FR-35: The system shall not attempt to identify or describe the specific modifications made to a tampered document; it shall only report that the content does not match the registered record.

FR-36: The system shall record every detected tampering event in the verification history log, including the verification ID, timestamp, and outcome.

4.5 Verification History

4.5.1 Description and Priority

This feature maintains a log of verification attempts performed through the system, supporting auditing and allowing authorized users to review past verification activity. Priority: Medium — this feature supports accountability and traceability but is not required for a single verification transaction to succeed.

4.5.2 Stimulus/Response Sequences

- A verification request completes (successfully or otherwise) → System creates a log entry containing the verification ID (if provided), a timestamp, the verification method used, and the result → System stores the entry in the verification history.
- A logged-in user requests to view their verification history → System retrieves and displays the log entries associated with that user's verification activity.
- An administrator requests to view system-wide verification activity → System retrieves and displays all verification log entries, with filtering options (e.g., by date range or result type).

4.5.3 Functional Requirements

FR-37: The system shall create a verification history entry for every verification attempt, regardless of outcome.

FR-38: Each verification history entry shall record, at minimum, the verification ID or file identifier used, the timestamp of the attempt, the verification method (ID, QR code, or file upload), and the result (Verified, Invalid, Not Found, or Tampered).

FR-39: The system shall allow a logged-in user to view a list of verification attempts they have personally performed.

FR-40: The system shall allow an Administrator to view all verification history entries across all users.

FR-41: The system shall prevent ordinary users (General Users and Issuing Organizations) from editing or deleting verification history entries.

FR-42: The system should allow an Administrator to filter verification history entries by date range and by result type.

4.6 Admin Management

4.6.1 Description and Priority

This feature provides system administrators with the tools needed to manage user accounts, oversee registered documents, and monitor system activity. Priority: Medium — required for ongoing operation and governance of the system, though not required for a single end-to-end verification transaction.

4.6.2 Stimulus/Response Sequences

- Administrator views the list of pending Issuing Organization registration requests → System displays the list → Administrator approves or rejects a request → System updates the account status accordingly.
- Administrator searches for a specific user account → System displays matching accounts → Administrator disables or re-enables the selected account.

- Administrator searches for a specific document record → System displays matching records → Administrator marks a record as invalid/revoked where appropriate.
- Administrator views system-wide verification activity → System displays a summary and detailed list of recent verification attempts.

4.6.3 Functional Requirements

FR-43: The system shall allow an Administrator to view a list of pending Issuing Organization account requests.

FR-44: The system shall allow an Administrator to approve or reject a pending Issuing Organization account request.

FR-45: The system shall allow an Administrator to search for and view details of any registered user account.

FR-46: The system shall allow an Administrator to disable a user account, preventing that account from logging in until re-enabled.

FR-47: The system shall allow an Administrator to search for and view details of any registered document record.

FR-48: The system shall allow an Administrator to mark a registered document record as invalid or revoked, after which the system shall return an "Invalid" result for that record on subsequent verification attempts.

FR-49: The system shall log all administrative actions that change the status of a user account or document record, including the administrator identity, action taken, and timestamp.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

NFR-01: The system shall return a verification result (for ID- or QR-code-based verification) within 3 seconds under normal load conditions (up to 50 concurrent users) on the reference deployment infrastructure.

NFR-02: The system shall complete document registration (excluding file upload transfer time) within 5 seconds under normal load conditions.

NFR-03: The system shall support at least 100 registered documents and 500 verification log entries without measurable degradation in verification response time, for the purposes of academic evaluation and demonstration.

5.2 Safety Requirements

The system does not control any physical equipment and does not pose direct physical safety risks. "Safety" in this context is interpreted as data-safety: preventing loss or corruption of stored records.

NFR-04: The system shall not permanently delete a document record or verification log entry as a result of a single user action; deletion/revocation of records shall be a reversible status change (e.g., "revoked") rather than physical deletion, except where an Administrator explicitly performs a permanent deletion.

NFR-05: The system shall validate all file uploads to ensure that only supported document file types are processed, reducing the risk of processing malformed or unsafe files.

5.3 Security Requirements

NFR-06: The system shall require authentication for all functionality other than document verification and account registration.

NFR-07: The system shall enforce role-based access control, ensuring that only users with the appropriate role (General User, Issuing Organization, or Administrator) can access the corresponding functionality.

NFR-08: The system shall store user passwords using a secure, salted hashing algorithm (e.g., bcrypt) and shall never store passwords in plain text or reversible encryption.

NFR-09: The system shall use HTTPS for all client-server communication to protect data in transit.

NFR-10: The system shall store uploaded documents in a manner that is not directly accessible via a public URL without passing through the application's access-control checks.

NFR-11: The system shall validate and sanitize all user input on the server side to reduce the risk of injection attacks (e.g., SQL injection, cross-site scripting).

NFR-12: The system shall maintain an audit log of administrative actions and verification attempts, as described in Sections 4.5 and 4.6.

NFR-13: The system shall limit the number of consecutive failed login attempts for a given account within a defined time window (default: 5 attempts per 15 minutes) to reduce the risk of brute-force attacks.

5.4 Software Quality Attributes

Reliability

NFR-14: The system shall correctly return a consistent verification result for the same verification ID and document content across repeated requests.

Availability

NFR-15: The system shall be available for use during normal demonstration and evaluation periods, targeting at least 95% uptime during any scheduled evaluation window.

Usability

NFR-16: A first-time user shall be able to complete a document verification (given a valid verification ID) without external assistance, within 2 minutes of first accessing the verification page.

Maintainability

NFR-17: The system's source code shall be organized into clearly separated modules (e.g., authentication, document registration, verification, administration) to support independent maintenance and testing of each module.

Scalability

NFR-18: The system's database schema and application logic shall support growth in the number of registered documents and users without requiring a redesign of the core data model.

Portability

NFR-19: The system's client interface shall function correctly on the latest two major versions of at least two widely used web browsers (e.g., Chrome and Firefox).

Testability

NFR-20: Each functional requirement listed in Section 4 shall be verifiable through a corresponding test case that can be executed manually or through automated testing.

5.5 Business Rules

- Only users with an approved Issuing Organization role may register or issue official documents.
- Only authenticated Administrators may approve Issuing Organization account requests or manage user accounts.
- A verification result shall always be generated based on stored document records; the system shall never display a "Verified" result for a document or ID that does not exist in its records.
- Invalid or non-existent verification IDs shall always return a "Not Found" result and shall never be displayed as "Verified".
- Verification history entries shall not be editable or deletable by General Users or Issuing Organization users.
- A document record that has been revoked by an Issuing Organization or Administrator shall return an "Invalid" result on all subsequent verification attempts, regardless of hash match.

6. Other Requirements

6.1 Database Requirements

- The system shall use a relational database to store user accounts, document metadata, integrity hashes, and verification history logs.
- The database schema shall enforce referential integrity between document records, the issuing-organization accounts that created them, and associated verification log entries.

6.2 Data Retention

- Verification history log entries shall be retained for a minimum period to be defined by institutional policy (a default retention period of 1 year is suggested for the academic prototype).
- Revoked or invalidated document records shall be retained (not physically deleted) so that verification attempts against them can still be correctly reported as "Invalid" rather than "Not Found".

6.3 Backup Requirements

- The database shall be backed up on a regular schedule (e.g., daily) to prevent loss of registered document and verification data. Exact backup frequency and retention are implementation/deployment decisions outside the direct scope of this SRS.

6.4 Legal and Privacy Considerations

The system stores personal and organizational data (account details, document metadata, and, where applicable, uploaded document files). This data shall be handled in accordance with applicable institutional data-handling guidelines. This SRS does not make any claim of formal legal or regulatory certification (e.g., specific data-protection law compliance); such certification, if required, is outside the scope of this academic project and would need to be separately assessed.

6.5 Localization / Internationalization

The initial release of the system is intended to support the English language only. The user interface text shall be organized (e.g., through the use of externalized string resources rather than hard-coded text) in a manner that would allow future localization to additional languages, although multi-language support is not a requirement for the initial release.

7. Appendices

Appendix A: Glossary

Term	Definition
SRS	Software Requirements Specification — a document that describes the intended functional and non-functional behavior of a software system.
Digital Document	An electronic file (e.g., a PDF) representing a certificate, letter, or record that is registered in and can be verified through the system.
Verification	The process of checking whether a submitted document or verification ID corresponds to an authentic, unmodified record stored in the system.
Authentication	The process of confirming the identity of a user, typically through a username/email and password.
Authorization	The process of determining whether an authenticated user has permission to perform a specific action or access specific functionality.
QR Code	A two-dimensional barcode that can encode a verification ID, allowing it to be scanned rather than typed manually.
Document ID / Verification ID	A unique identifier generated by the system when a document is registered, used to look up the corresponding record during verification.
Hash / Integrity Check	A fixed-length value computed from a document's content using a cryptographic hash function (e.g., SHA-256); used to detect whether the content has changed since registration.
Administrator	A system user responsible for managing user accounts, document records, and monitoring system activity.
Issuing Organization	An authorized organization permitted to register/issue documents within the system.
API	Application Programming Interface — a defined set of endpoints through which the front-end and other clients communicate with the server-side application.
Database	The relational data store used by the system to persist user, document, and verification-log data.

Appendix B: Analysis Models

The following diagrams are recommended to support the design phase of this project. Each diagram should be created separately using an appropriate modeling tool; their intended content is described below.

Use Case Diagram

Should show the three actors (General User/Verifier, Issuing Organization, Administrator) and their associated use cases, including Register Account, Log In, Register Document, Verify Document, View

Verification History, Approve Issuing Organization, Manage Users, and Manage Documents. Include relevant «include»/«extend» relationships (e.g., "Verify Document" may include "Check Integrity Hash").

Data Flow Diagram (Level 0 and Level 1)

The Level 0 (context) diagram should show the system as a single process with external entities (Verifier, Issuing Organization, Administrator) and the data flows between them (e.g., verification request, verification result, document registration data). The Level 1 diagram should decompose the system into major processes such as Manage Accounts, Register Document, Verify Document, and Maintain Verification Log, along with the data stores they read from and write to (User Store, Document Store, Verification Log Store).

ER Diagram

Should show entities such as User (with a role attribute), Document (linked to the Issuing Organization that registered it, containing metadata and the integrity hash), and VerificationLog (linked to the Document and, where applicable, the User who performed the verification), along with their attributes and relationships (e.g., one Issuing Organization registers many Documents; one Document has many VerificationLog entries).

Class Diagram

Should show major system classes such as User, IssuingOrganization, Administrator (as specializations or role attributes of User), Document, VerificationRequest, and VerificationLog, along with their key attributes, methods (e.g., register(), verify(), computeHash()), and relationships (association, inheritance where applicable).

Activity Diagram

Should illustrate the step-by-step flow of the document verification process: start → user submits ID/QR/file → system looks up the record → decision (record found?) → if not found, display "Not Found" and end; if found, decision (hash matches?) → if not, display "Tampered"; if yes, decision (record active?) → display "Invalid" if revoked, otherwise display "Verified" → end.

Sequence Diagram

Should show the interaction between the Verifier (actor), the Web Client, the Application Server, and the Database during a document verification request: the client sends the verification request to the server; the server queries the database for a matching record; the database returns the record (or indicates none found); the server computes/compares the hash if a file was submitted; the server returns the verification result to the client for display.

Appendix C: To Be Determined List

The following items are intentionally left open and should be resolved during the design phase, based on the specific implementation platform and institutional context chosen for the project:

TBD ID	Item	Related Section(s)
TBD-01	Final choice of backend framework and database technology.	2.4, 2.5, 3.3

TBD ID	Item	Related Section(s)
TBD-02	Exact session-inactivity timeout value and whether it is user-configurable.	4.1 (FR-11)
TBD-03	Whether unauthenticated (guest) users are permitted to perform document verification, or whether login is required for all verification requests.	4.3 (FR-30)
TBD-04	Final list of supported document file types beyond PDF, if any.	4.2 (FR-20)
TBD-05	Exact data retention period for verification logs, to be confirmed with institutional/project guide policy.	6.2